Diameter-7 cells from a spread and its whiskers: the freed sets and the bounds for every q

Degree–diameter lab, 29 October 2026 (job thm-conics)

Abstract. The jobs "conics" and "conics-2" raised the lab's values at the diameter-7 cells $(14,7)$ to $(20,7)$ by freeing, on the friendly transitions of the typed hexagon layouts, two sets: the points of the Hermitian spread S of $H(q)$, and the lines meeting a spread line v together with an exact whisker set and the q^2 spread lines that carry no whisker. Here these steps are turned into statements for every prime power q. (1) For any generalized hexagon of order (q,q), any spread S, any $v \in S$ and any whisker set W, the lines meeting v, the whiskers and the spread lines missed by the whiskers form a dual segment set; for the exact whisker set of `POLISHED.md` (every odd q and every even $q \geq 4$) it has exactly q^4+q^2+q lines and is maximal, while the spread points form a segment set of $(q+1)(q^3+1)$ points. (2) For $2k+1$ levels, any regular tournament and any typing, some level has at most $\lfloor k/2 \rfloor$ friendly outgoing transitions and some level at most $\lfloor k/2 \rfloor$ friendly incoming ones; this bounds the hub counts, with the exact second-order term, and the alternating typing attains the bound for the total number of hubs for every k. (3) With the q^2+1 points of an elliptic quadric section on the non-friendly transitions, this gives, for every $k \geq 2$ and every prime power $q \geq q_k$,

$$n_b(q+1+k,7) \;\geq\; 2(2k+1)(q+1)(q^4+q^2+1) + \begin{cases} k\,(q^4+2q^2+q+1) & k \text{ even}, \\ \frac{k-1}{2}\,(2q^4+q^3+q^2+2q+1)+(k+1)(q^2+1) & k \text{ odd}, \end{cases}$$

with $q_2 = 16$, $q_3 = q_4 = 23$, $q_5 = 29$, $q_k \leq 300k$ in general. The hub graphs are given by a new explicit construction (sums of blown-up hexagons); smaller q are checked by computer where possible. The formulas reproduce the vertex counts of all fifteen graphs built in the two jobs. At $(19,7)$ the formula equals the built value 11,316,940 and every ingredient is proved. At $(18,7)$ the theorem with an explicit hub graph (two copies of the incidence graph of $H(7)$ joined by perfect matchings) gives 7,318,644, above the lab's value 7,309,092; that graph was built, and checked from every hub (§5.1).

Labels. Every result is marked **Theorem** (proved here or in the cited note), **Computer-checked** (a program checked it; the program is in the repository) or **Heuristic**, and **[new]** or **[restatement]**. "New" means new in the lab's notes; the literature check of `D7_HEXAGON.md` §11 applies, and we did not search the literature again.

Sources. `research/synthetic/CONICS.md` §3.4 (branch `claude/conics-generalized-polygons-6gjqxi`); `research/synthetic/conics2/README.md` (branch `claude/conics-2-diameter-7-ic0oat`, state of 11:44 UTC); `research/theorems/D7_HEXAGON.md` (branch `claude/d7-hexagon-theorems-4w1prt`), whose Definitions 2.1, 3.1, Theorems A, B, C, 3.4 and Proposition 5.2 are used throughout; `research/theorems/POLISHED.md` §6 (branch `claude/thm-polish-2oddll`); `research/synthetic/SYNTHETIC.md` (branch `claude/synthetic-incidence-geometry-s4m08f`), whose conventions we follow. Programs of this note: `research/synthetic/d7thm/`; squad round `research/squads/d7thm-r1/`.

0. Setting

Hexagons. A generalized hexagon Γ of order (q,q) has $N = N(q) = (q+1)(q^4+q^2+1)$ points and as many lines; $q+1$ points on a line and $q+1$ lines on a point. Distances are taken in the incidence graph I (diameter 6, girth 12); elements at distance 6 are *opposite*. For an element x, $|\Gamma_i(x)| = (q+1)q^{i-1}$ for $1 \leq i \leq 5$. A *spread* is a set of q^3+1 pairwise opposite lines; every line not in a spread S meets exactly one line of S, in one point (`D7_HEXAGON.md`, Lemma 1.2(c)). The example throughout is the split Cayley hexagon $H(q)$ in $Q(6,q)$ and its Hermitian spread: the lines of $H(q)$ inside a hyperplane Π with $\Pi \cap Q(6,q)$ elliptic (every q; [Th80]).

Segment sets (D7_HEXAGON.md, Definition 1.3). A point set F is a *segment set* if for every $p \in F$ the points of $F \setminus \{p\}$ collinear with p lie on one line through p. A line set G is a *dual segment set* if for every $\ell \in G$ the lines of $G \setminus \{\ell\}$ that meet ℓ pass through one point of ℓ. Both kinds have at most $(q+1)(q^3+1)$ elements (D7_HEXAGON.md, Proposition 1.4).

Layouts. $L = 2k+1$ *levels*, each a copy of I of *type* H (colour $\mathcal{A}$ = points, $\mathcal{B}$ = lines) or D (colour $\mathcal{A}$ = lines, $\mathcal{B}$ = points). The levels are the vertices of a *regular tournament* T: every two levels are joined by exactly one *transition* $j \to j'$, and every level has k outgoing and k incoming transitions. A transition is a bijection from the $\mathcal{A}$-elements of level j to the $\mathcal{B}$-elements of level j'; it is *friendly* if the types differ, and then it is the identity. D7_HEXAGON.md uses the rotational tournament $j \to j+i$ ($1 \le i \le k$, indices mod L); Theorem A and Theorem 3.4 there hold verbatim for every regular tournament, since their proofs only use that every pair of levels is the pair of one transition. The *freed set* R_t of a transition is the set of $\mathcal{A}$-elements whose matching edge is deleted. Level j then has $\alpha_j = \sum_{t \text{ out of } j} |R_t|$ free $\mathcal{A}$-slots and $\beta_j = \sum_{t \text{ into } j} |R_t|$ free $\mathcal{B}$-slots. $K_{\mathcal{B}} \le \min_j \alpha_j$ hubs of colour $\mathcal{B}$ each take one free $\mathcal{A}$-slot in every level, $K_{\mathcal{A}} \le \min_j \beta_j$ hubs of colour $\mathcal{A}$ one free $\mathcal{B}$-slot in every level, and a bipartite *hub graph* Y joins the two kinds of hubs.

What repairs a transition. Theorem B of D7_HEXAGON.md: a friendly transition is repaired if its freed set is a segment set (source of type H) or a dual segment set (source of type D). Theorem C there, with Corollary 5.3(a): a non-friendly transition $H \to H$ with matching σ is repaired by a set R of m pairwise opposite points when σ maps R onto m pairwise opposite lines and the points collinear with x onto the lines concurrent with $\sigma(x)$ for every $x \in R$; such a σ exists for every partial ovoid R and every partial spread of the same size. A $D \to D$ transition uses σ^{-1} and frees $\sigma(R)$.

Hub graphs. $\mathrm{HG}(K_1, K_2; \delta)$ denotes a bipartite graph with colour classes of sizes K_1 and K_2, maximum degree at most δ, in which any two vertices *of the same colour* are at distance at most 6.

Theorem 0.1 (D7_HEXAGON.md, Theorem 3.4) [restatement]. If every transition is repaired, $K_{\mathcal{B}} \le \min \alpha_j$, $K_{\mathcal{A}} \le \min \beta_j$ and the hub graph is an $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; q-k)$, then the graph has maximum degree $q+1+k$ and diameter at most 7, and

$$n_b(q+1+k, 7) \;\ge\; 2(2k+1)N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

The value $2(2k+1)N(q)$ alone (no freed sets, no hubs) is the value of the 2013 table at the cells $(7,7)$, $(9,7)$–$(16,7)$ [MS13], and the lab's beyond-table value at $(17,7)$–$(20,7)$.

What uses coordinates. Sections 2 and 4 are about tournaments and bipartite graphs only. In §1, Theorem 1.4, Proposition 1.5 and Lemma 1.6 hold in every generalized hexagon of order (q,q) with a spread; only Theorem 1.7 (the whisker set of POLISHED.md meets each spread line in 0 or $q+1$ points, so exactly q^2 spread lines are missed) and the existence statements (Hermitian spread, exact whisker set, elliptic quadric section) use the coordinates of $H(q)$. Coordinates appear only in proofs.

Notation for sizes. $P(q) = (q+1)(q^3+1) = q^4+q^3+q+1$, $\Lambda(q) = q^4+q^2+q$, and m the size of the non-friendly sets. Note $P - \Lambda = q^3 - q^2 + 1$.

1. The freed sets

1.1 The configuration

Definition 1.1. Let S be a spread of Γ, $\mathcal{P}_S$ the set of points on its lines, and $v \in S$. $\mathcal{L}_v$ is the set of the $q(q+1)$ lines other than v that meet v. A point is *free* if it is at distance 5 from v and not in $\mathcal{P}_S$. A *whisker set* is a set W of lines, each opposite v and not in S, such that no free point lies on two lines of W; it is *exact* if every free point lies on exactly one line of W. Finally

$$S_0(W) = \{n \in S \setminus \{v\} : \; n \text{ meets no line of } W\}, \qquad R_L(W) = \mathcal{L}_v \cup W \cup S_0(W).$$

(This is the configuration of Theorem D, POLISHED.md §6.2 and SYNTHETIC.md §3.2, written without coordinates.)

Lemma 1.2 [restatement, with the counts; Theorem]. (a) The points at distance at most 3 from v are the $q+1$ points of v and the $q^2(q+1)$ points at distance 3, each of which lies on exactly one line of $\mathcal{L}_v$. (b) The points of $\mathcal{P}_S$ not on v are at distance 5 from v. (c) There are $q^3(q^2-1)$ free points. (d) A line opposite v and not in

S has exactly one point in $\mathcal{P}_S$ and q free points. (e) A whisker set has at most $q^2(q^2-1)$ lines, with equality if and only if it is exact.

Proof. (a) A point at distance 3 from v lies on a line meeting v; two such lines through it would close a circuit of length 4 or 6. Counting: $|\Gamma_3(v)| = (q+1)q^2$. (b) A point x on a spread line $n \neq v$ has $d(x,v)$ odd, at most 5 and at least $d(n,v)-1 = 5$. (c) $|\Gamma_5(v)| = (q+1)q^4$, of which $q^3(q+1)$ lie in $\mathcal{P}_S$ by (b). (d) Every point of a line opposite v is at distance 5 from v; the line meets exactly one line of S, in one point. (e) By (d) and (c), counting incidences between free points and W. □

1.2 Spread points and whiskers

Proposition 1.3 (`D7_HEXAGON.md`, Proposition 1.4) [restatement; Theorem]. $\mathcal{P}_S$ is a segment set of $P(q) = (q+1)(q^3+1)$ points, and no segment set is larger; equality holds only for the point sets of spreads.

Theorem 1.4 [new in this form; Theorem]. Let S be a spread, $v \in S$ and W any whisker set.

(a) $R_L(W) = \mathcal{L}_v \cup W \cup S_0(W)$ is a dual segment set, and two of its lines that meet, meet in a point of $\mathcal{P}_S$.

(b) $|R_L(W)| = q(q+1) + |W| + |S_0(W)|$.

(c) If W is exact and every point of $\mathcal{P}_S$ not on v lies on a line of W or of $S_0(W)$, then every point of Γ lies on a line of $R_L(W)$, and no line can be added to $R_L(W)$: it is a maximal dual segment set.

(d) A line $n \in S \setminus \{v\}$ all of whose whiskers pass through one point of n can be added to $R_L(W)$ as well.

Proof. (a) A line of $\mathcal{L}_v$ has one point on $v \subseteq \mathcal{P}_S$ and its other q points at distance 3 from v, so not in $\mathcal{P}_S$ (Lemma 1.2(b)). A whisker has one point in $\mathcal{P}_S$ and q free points (Lemma 1.2(d)). So every line of $\mathcal{L}_v \cup W$ has exactly one point in $\mathcal{P}_S$. Its other points are either at distance 3 from v, and then on no other line of $\mathcal{L}_v$ (Lemma 1.2(a)) and on no whisker (whose points are at distance 5), or free, and then on no line of $\mathcal{L}_v$ and on at most one whisker. So two lines of $\mathcal{L}_v \cup W$ never meet outside $\mathcal{P}_S$. A line $n \in S_0(W)$ meets no line of $\mathcal{L}_v$ (whose points are at distance at most 3 from v, while those of n are at distance 5), no whisker (definition) and no other spread line. So the lines of $S_0(W)$ are isolated, and for every other line ℓ of $R_L(W)$ the lines of $R_L(W)$ meeting ℓ all pass through the unique point of ℓ in $\mathcal{P}_S$. (b) The three sets are disjoint: whiskers are opposite v, lines of $\mathcal{L}_v$ are not, and neither kind is in S. (c) Points of v and points at distance 3 from v lie on lines of $\mathcal{L}_v$, free points on whiskers (W exact), and the points of $\mathcal{P}_S$ off v by hypothesis. A line $\ell \notin R_L(W)$ then meets a line of $R_L(W)$ at each of its $q+1$ points, and these lines are different from each other (two points of ℓ on one line $\neq \ell$ is impossible), so $R_L(W) \cup \{\ell\}$ is not a dual segment set. (d) Such an n meets lines of $R_L(W)$ only in that point x (it meets no line of $\mathcal{L}_v$ and no other spread line), and for a whisker through x the new line meets it at its point in $\mathcal{P}_S$, where all its other meetings already are. Two such spread lines do not meet. □

For W exact and $S_0(W)$ left out this is `CONICS.md`, Lemma 3.1; for the whisker set of `POLISHED.md` and its q^2 empty spread lines it is Proposition 1(c),(d) of `conics2/README.md`. The proof above uses only incidences.

Proposition 1.5 [new; Theorem] (*where the room is*). Let G be a dual segment set in which any two lines that meet do so in a point of $\mathcal{P}_S$ (as in Theorem 1.4(a)). Then $|G \setminus S| \leq q^4+q$, with equality if and only if the lines of $G \setminus S$ cover every point outside $\mathcal{P}_S$ exactly once. Hence $|G| \leq q^4+q+|G \cap S|$. For an exact whisker set, $\mathcal{L}_v \cup W$ has exactly q^4+q lines.

Proof. A line not in S meets exactly one spread line, in one point, so it has exactly one point in $\mathcal{P}_S$ (a second point would lie on a second spread line). Its other q points lie outside $\mathcal{P}_S$, and by hypothesis these q-sets are disjoint for different lines of $G \setminus S$. There are $N - P(q) = q^5 + q^2 = q(q^4+q)$ points outside $\mathcal{P}_S$. Finally $q(q+1) + q^2(q^2-1) = q^4+q$. □

So within this class $R_L(W)$ can only grow by containing more spread lines, at most q^3-q^2+1 more; the general bound $(q+1)(q^3+1) = q^4+q+(q^3+1)$ is the case where all spread lines are in G, which forces an ovoid.

Lemma 1.6 [new; Theorem]. Let W be exact. Every whisker meets exactly one line of $S \setminus \{v\}$, and a line $n \in S \setminus \{v\}$ meets at most q^2-1 whiskers. So at least q^2 spread lines other than v meet a whisker and $|S_0(W)| \leq q^3-q^2$. If every line of $S \setminus \{v\}$ meets either no whisker or $q+1$ whiskers, then $|S_0(W)| = q^2$.

Proof. A whisker meets one line of S (§0), not v, since it is opposite v. At a point x of n the $q+1$ lines are n, the line through x on the geodesic to v (at distance 4 from v, so not a whisker) and $q-1$ others. Hence n meets

at most $(q+1)(q-1)$ whiskers, and $q^2(q^2-1)/(q^2-1) = q^2$. In the last case the lines meeting whiskers number $q^2(q^2-1)/(q+1) = q^2(q-1)$, and $q^3 - q^2(q-1) = q^2$. □

1.3 The exact whisker set of `POLISHED.md`

In $H(q)$ with the coordinates of `POLISHED.md` §6.1 (q odd) and §6.5 (q even): Π the elliptic hyperplane $X_0 + cX_4 = 0$ (q odd, $-c$ a non-square) or $X_3 = X_0 + \delta X_4$ (q even, $\operatorname{Tr}\delta = 1$), S its Hermitian spread, $v = \langle e_1, e_6\rangle$, the hyperplanes $\Pi_z \supset v$ ($z = (\alpha, \beta) \in \mathbb{F}^2$) and their spreads S_z, the function ν on the points of Π off v, the zero-free cubic G and $F(z) = G(z)/N(z)$ with N the anisotropic norm form; $W = \{\ell \in S_z \ \ \{v\} : z \neq 0,\ \nu(\ell \cap \Pi) = F(z)\}$ is exact for every odd q (Theorem 6.5) and every even $q \geq 4$ (Theorem 6.14).

Theorem 1.7 (`conics2/README.md`, Proposition 1(a),(b)) [restatement with proof; Theorem]. Let q be odd, or even with $q \geq 4$, and W the exact whisker set above. Then every point X of $\mathcal{P}_S$ not on v lies on exactly one whisker if $\nu(X) \neq 0$ and on none if $\nu(X) = 0$; ν is constant on every line of $S \ \ \{v\}$. Hence every line of $S \ \ \{v\}$ meets either $q+1$ whiskers, one at each of its points, or none, and $S_0(W) = \{n \in S \ \ \{v\} : \nu(n) = 0\}$ has exactly q^2 lines.

Proof. Whiskers through X are whiskers whose point in Π is X (their other points are off Π). On Π the equation of Π_z reduces to $\alpha X_2 + \beta X_5 = 0$ (both parities), and $(X_2, X_5) \neq (0,0)$ because X is not on v (`POLISHED.md`, Lemma 6.1(d) / 6.10). So the $z \neq 0$ with $X \in \Pi_z$ form the punctured line $\{tz_0 : t \neq 0\}$, $z_0 = (X_5, -X_2)$. For each of them exactly one line of S_z passes through X (Lemma 6.1(b); for even q read Lemma 6.10 for 6.1 throughout); it is not v, it is not in S (6.1(e)), so it meets Π only in X (6.1(c)), and different z give different lines (6.1(e)). It is a whisker if and only if $\nu(X) = F(tz_0)$. As G is a cubic form and N a quadratic form, $F(tz_0) = tF(z_0)$, and $F(z_0) \neq 0$ since G has no zero. So $t \mapsto F(tz_0)$ is a bijection of $\mathbb{F}^\times$: one whisker through X if $\nu(X) \neq 0$, none if $\nu(X) = 0$. ν is constant on spread lines by `POLISHED.md`, Lemma 6.6(d) / 6.12(d). The count is Lemma 1.6. □

Corollary 1.8 [Theorem]. For every prime power q with q odd or $q \geq 4$, in $H(q)$: $R_H = \mathcal{P}_S$ is a segment set with $P(q) = q^4 + q^3 + q + 1$ points, and $R_L = \mathcal{L}_v \cup W \cup S_0(W)$ is a maximal dual segment set with $\Lambda(q) = q^4 + q^2 + q$ lines (Theorem 1.4 with Theorem 1.7). In the notation of Proposition 1.5, $\mathcal{L}_v \cup W$ is extremal and $|R_L \cap S| = q^2$.

Computer-checked. `conics2/typed_build.py` asserts every statement of Theorem 1.7 and Corollary 1.8 for $q = 4, 5, 7, 11, 13, 16$ inside the builds. Independently, squad round `d7thm-r1` (own code, not the builder): card c1 for $q = 3, 5, 7$ (hexagon built from the quadric and Tits' equations), card c2 for $q = 9, 11$, card c3 for $q = 2, 4, 8$: exactness of W, ν constant on spread lines, 0 or $q+1$ whiskers per spread line, $|S_0| = q^2$ equal to the lines with $\nu = 0$, the dual segment property, $|R_L| = q^4 + q^2 + q$ (6,651 at $q = 9$, 14,773 at $q = 11$, 276 at $q = 4$, 4,168 at $q = 8$), maximality (every outside line tested), and the segment property of $\mathcal{P}_S$; card c2 also for every admissible c at $q = 9, 11$, card c3 for every admissible (c_0, c_1) at $q = 4, 8$; negative controls fail as they should (`research/squads/d7thm-r1/scoreboard.jsonl`).

Remarks. (i) $q = 2$: Lemma 6.11 of `POLISHED.md` gives no zero-free G (card c3 re-checked all four pairs), but an exact whisker set of 12 lines exists (`POLISHED.md`, Corollary 6.15), so Theorem 1.4 gives $|R_L| \geq 18$; the largest dual segment set of $H(2)$ has 25 lines (**Computer-checked**, `conics2/README.md` §3: SAT and MaxSAT, squad round `conics2-r1` cards c6, c7). (ii) R_L is not maximum in general: $H(4)$ has a dual segment set of 288 $>$ 276 lines (**Computer-checked**, same place: card c8, re-checked by `research/squads/conics2-r1/tools/dualseg_check.py`; it is not of the kind in Proposition 1.5). Whether $q^4 + q^2 + q$ can be beaten for $q \geq 5$ is open; by Proposition 1.5 a set of this kind would need more than q^2 spread lines. (iii) For $q = 3^h$ the pencils of an ovoid give a dual segment set of $P(q)$ lines, and the duality layout of `D7_HEXAGON.md`, Corollary 8.2 (every freed set of size $P(q)$) is at least as good as any typed layout, by Theorem 2.1. (iv) We know no coordinate-free description of the q^2 lines with $\nu = 0$.

1.4 The non-friendly sets

Proposition 1.9 (`D7_HEXAGON.md`, Lemma 1.2(a), Corollaries 5.3(a) and 9.1) [restatement; Theorem]. For every prime power q, the $q^2 + 1$ points of an elliptic quadric section $Q^-(3, q)$ of $Q(6, q)$ are pairwise opposite in $H(q)$, and with any $q^2 + 1$ lines of the Hermitian spread they give a local duality σ as in Theorem C. So $m = q^2 + 1$ is available on every non-friendly transition.

The lab's builds use larger non-friendly sets found by computer at $q = 11$ ($m = 768$, mirrored set) and $q = 13$ ($m = 792$), and $m = q^2 + 1 = 257$ at $q = 16$ (**Computer-checked**: the builder re-checks non-collinearity,

non-concurrency and the local duality at every point).

2. Slot counts

Let T be a regular tournament on $2k+1$ levels with a typing. For a level j let f_j be the number of friendly transitions out of j and g_j the number into j. With friendly sets of sizes P (points, source H) and Λ (lines, source D) and non-friendly sets of size m,

$$\alpha_j = f_j X_j + (k-f_j)m, \quad X_j = \begin{cases} P & \tau_j = H \\ \Lambda & \tau_j = D \end{cases}, \qquad \beta_j = g_j Y_j + (k-g_j)m, \quad Y_j = \begin{cases} \Lambda & \tau_j = H \\ P & \tau_j = D \end{cases}$$

(a friendly transition into an H level comes from a D level, which frees lines). Put $a = \lfloor k/2 \rfloor$ and assume

$$P \;\geq\; \Lambda \;\geq\; m, \qquad \Lambda - m \;\geq\; a\,(P-\Lambda). \tag{$*$}$$

For the sets of §1, $(*)$ holds whenever $k \leq 2q+1$ and $m \leq q^2+1$, since then $a(q^3-q^2+1) \leq q^4-q^3+q \leq q^4+q-1$; at $q=11$, $m=768$ it holds for $k \leq 23$, at $q=13$, $m=792$ for $k \leq 27$. The bound needs $k \leq q-1$ anyway (hub degree $q-k \geq 1$).

Theorem 2.1 [new; Theorem]. For every regular tournament on $2k+1$ levels and every typing, with $a = \lfloor k/2 \rfloor$:

(a) $\min_j f_j \leq a$ and $\min_j g_j \leq a$;

(b) if k is even, it is impossible that $f_j \geq a$ at every level and $f_j \geq a+1$ at every D level; likewise with g and H levels;

(c) if k is odd, it is impossible that $f_j \geq a+1$ at every D level and $g_j \geq a+1$ at every H level.

Consequently, if $P \geq \Lambda \geq m$: for k odd, $\min\alpha, \min\beta \leq aP+(a+1)m$ and $\min\alpha + \min\beta \leq a(P+\Lambda)+2(a+1)m$; for k even and under $(*)$, $\min\alpha \leq a(\Lambda+m)$ and $\min\beta \leq a(\Lambda+m)$.

Proof. Let h and d be the numbers of H and D levels. The $k-f_j$ non-friendly transitions out of an H level go to H levels, so $\sum_{\tau_j=H}(k-f_j) = \binom{h}{2}$, and likewise $\sum_{\tau_j=D}(k-f_j) = \binom{d}{2}$ and the same with g.

(a) If $f_j \geq a+1$ everywhere, then $\binom{h}{2} \leq h(k-a-1)$ and $\binom{d}{2} \leq d(k-a-1)$, so $h+d \leq 4k-4a-2 \leq 2k$, less than $2k+1$. For g the same.

(b) $k=2a$. From $f_j \geq a$ on H levels, $\binom{h}{2} \leq ha$, so $h \leq 2a+1$; from $f_j \geq a+1$ on D levels, $\binom{d}{2} \leq d(a-1)$, so $d \leq 2a-1$. Then $h+d \leq 4a < 4a+1$. For g exchange the roles of H and D.

(c) $k=2a+1$. On H levels $g_j \geq a+1$ gives $\binom{h}{2} \leq ha$ (count non-friendly incoming transitions), so $h \leq 2a+1$; on D levels $f_j \geq a+1$ gives $\binom{d}{2} \leq da$, so $d \leq 2a+1$. Then $h+d \leq 4a+2 < 4a+3$.

Consequences. A level with $f_j \leq a-1$ has $\alpha_j \leq (a-1)P+(a+1)m \leq a\Lambda+am$, by $(*)$ (only $\Lambda-m \geq (a-1)(P-\Lambda)$ is used, which always holds for $k=2$). If k is even and $\min\alpha > a(\Lambda+m)$, every level has $f_j \geq a$, and every D level $f_j \geq a+1$ (a D level with $f_j = a$ has $\alpha_j = a\Lambda+am$), contradicting (b). If k is odd, a level with $f_j \leq a$ has $\alpha_j \leq aP+(a+1)m$, so by (a) $\min\alpha \leq aP+(a+1)m$, and the same for β. If $\min\alpha+\min\beta > a(P+\Lambda)+2(a+1)m$, both minima exceed $a\Lambda+(a+1)m$. A D level with $f_j \leq a$ has $\alpha_j \leq a\Lambda+(a+1)m$ and an H level with $g_j \leq a$ has $\beta_j \leq a\Lambda+(a+1)m$, so every D level has $f_j \geq a+1$ and every H level $g_j \geq a+1$, contradicting (c). □

Theorem 2.2 [new; Theorem] (*the alternating layout*). Take the rotational tournament $j \to j+i$ $(1 \leq i \leq k)$ on $\mathbb{Z}_{2k+1}$ and the typing $\tau_j = H$ for j even, $\tau_j = D$ for j odd $(0 \leq j \leq 2k$; the levels $2k$ and 0 are both of type $H)$. Then $f_j = \lceil k/2 \rceil$ for $0 \leq j \leq k$, and for $j = k+t$ $(1 \leq t \leq k)$

$$f_j = \begin{cases} k/2 \text{ (t even, H)} \quad k/2+1 \text{ (t odd, D)} & k \text{ even}, \\ (k-1)/2 \text{ (t odd, H)} \quad (k+1)/2 \text{ (t even, D)} & k \text{ odd}; \end{cases}$$

and $g_j = f_{2k-j}$. Hence (for odd k using $(*)$)

$$k \text{ even:} \quad \min\alpha = \min\beta = \tfrac{k}{2}(\Lambda+m); \qquad k \text{ odd:} \quad \min\alpha = \tfrac{k-1}{2}P + \tfrac{k+1}{2}m, \quad \min\beta = \tfrac{k-1}{2}\Lambda + \tfrac{k+1}{2}m,$$

For k even both minima attain the bounds of Theorem 2.1; for k odd $\min\alpha$ and $\min\alpha+\min\beta$ do.

Proof. For $j\le k$ the targets $j+1,\dots,j+k$ need no reduction mod $2k+1$, and exactly $\lceil k/2\rceil$ of them have the parity opposite to j. For $j=k+t$ the targets are $j+1,\dots,2k$ ($k-t$ of them, $\lceil(k-t)/2\rceil$ of the other parity) and $0,\dots,t-1$, of which $\lfloor t/2\rfloor$ (if j is even) or $\lceil t/2\rceil$ (if j is odd) have the other parity; the four cases give the values stated. The reflection $j\mapsto 2k-j$ preserves types and reverses every transition, so $g_j=f_{2k-j}$. For k even the minimum a of f is attained at D levels (odd $j\le k$), so $\min\alpha=a\Lambda+am$; the minimum of g is attained at H levels, where friendly incoming sets are lines: $\min\beta=a\Lambda+am$. For k odd, $f_j=a$ only at H levels and every D level has $f_j=a+1$, whose $\alpha_j=(a+1)\Lambda+am\ge aP+(a+1)m$ by $(*)$; so $\min\alpha=aP+(a+1)m$; $g_j=a$ only at H levels, where $\beta_j=a\Lambda+(a+1)m$. □

So, with friendly point sets of size P, friendly line sets of size Λ and non-friendly sets of size m, the alternating layout maximizes $K_{\mathcal{A}}+K_{\mathcal{B}}$ (and for k even each of them) over all regular tournaments and typings. Its slot counts are

$$K_{\mathcal{B}}(q,k)=K_{\mathcal{A}}(q,k)=\tfrac{k}{2}(\Lambda+m)\ (k\text{ even}),\qquad K_{\mathcal{B}}=\tfrac{k-1}{2}P+\tfrac{k+1}{2}m,\quad K_{\mathcal{A}}=\tfrac{k-1}{2}\Lambda+\tfrac{k+1}{2}m\ (k\text{ odd}).$$

For $k=1$ this is m on both sides, `D7_HEXAGON.md` Corollary 9.1.

The layouts of the builds [Computer-checked against the builders' logs]. The free slots per level, in level order:

k	types	α_j	β_j
2	$HDHDH$ (alternating)	$P{+}m$, $\Lambda{+}m$, $P{+}m$, 2Λ, $P{+}m$	$\Lambda{+}m$, $2P$, $\Lambda{+}m$, $P{+}m$, $\Lambda{+}m$
3	$HHDHHDD$	$P{+}2m$, $P{+}2m$, $2\Lambda{+}m$, $2P{+}m$, $2P{+}m$, $2\Lambda{+}m$, $2\Lambda{+}m$	$2\Lambda{+}m$, $2\Lambda{+}m$, $2P{+}m$, $\Lambda{+}2m$, $\Lambda{+}2m$, $2P{+}m$, $2P{+}m$
4	$HHDHDHDHD$ (alternating, rotated)	$2P{+}2m$, $2P{+}2m$, $2\Lambda{+}2m$, $2P{+}2m$, $2\Lambda{+}2m$, $2P{+}2m$, $3\Lambda{+}m$, $2P{+}2m$, $3\Lambda{+}m$	$2\Lambda{+}2m$, $2\Lambda{+}2m$, $3P{+}m$, $2\Lambda{+}2m$, $3P{+}m$, $2\Lambda{+}2m$, $2P{+}2m$, $2\Lambda{+}2m$, $2P{+}2m$

The $k=3$ layout of the builds is not the alternating one but has the same minima. With the values of each build these vectors equal, entry by entry, the vectors printed by `typed_build.py` (`d7thm/cells.out`).

Computer-checked. `d7thm/slots.py`: for $k\le 8$ every typing of the rotational tournament (the optimum is the one of Theorem 2.2, numerically for $q=11$ with $m=768$, $q=13$ with $m=792$, $q=16,101,1009$); every labelled regular tournament on 5 and 7 levels (24 and 2,640) with every typing; SAT (CaDiCaL) on $9,11,13$ levels finds no regular tournament and typing with $f_j,g_j\ge a$ everywhere, $f_j\ge a+1$ on D levels and $g_j\ge a+1$ on H levels (one combined instance; (b) and (c) themselves are proved above). Independently squad card c6: all regular tournaments on $3,5,7$ levels, 200,000 random ones on 9 and 11 levels, annealing, and the closed forms of Theorem 2.2 for $k\le 40$. Card c6 also found that without $(*)$ the value of $\min\alpha$ for odd k changes (the D levels become the minimum); this happens only for k far above q (for instance $k\ge 27$ at $q=11$), outside the range of the bound.

3. The bound

Theorem 3.1 [new; Theorem]. Let q be a prime power, q odd or $q\ge 4$, let $1\le k\le q-1$, and let $K_{\mathcal{B}}(q,k)$, $K_{\mathcal{A}}(q,k)$ be the slot counts of Theorem 2.2 with $P=P(q)$, $\Lambda=\Lambda(q)$ and $m=q^2+1$. If an $\mathrm{HG}(K_1,K_2;q-k)$ exists with $K_1\le K_{\mathcal{B}}$ and $K_2\le K_{\mathcal{A}}$, then

$$n_b(q+1+k,\,7)\ \ge\ 2(2k+1)N(q)+K_1+K_2.$$

Proof. Take $2k+1$ copies of the incidence graph of $H(q)$, the alternating layout of Theorem 2.2, the identity on friendly transitions with freed sets $R_H=\mathcal{P}_S$ (source H) and R_L (source D) of Corollary 1.8, and on non-friendly transitions the local duality σ ($H\to H$) or σ^{-1} ($D\to D$) of Proposition 1.9 with the q^2+1 freed elements. Every

friendly transition is repaired by Theorem B, every other one by Theorem C; $(*)$ holds since $k \le q-1$ (§2), so Theorem 2.2 gives the slot counts. Attach K_1 hubs of colour $\mathcal{B}$ and K_2 of colour $\mathcal{A}$ to free slots in any way (one slot per level each; unused slots stay free) and use the given graph as hub graph. Theorem 0.1 applies. □

The same holds with m replaced by the size of any non-friendly set satisfying Theorem C, provided $(*)$ holds; and for any generalized hexagon of order (q,q) with a spread, an exact whisker set with the property of Theorem 1.7, and a partial ovoid of a size m satisfying $(*)$ (for instance $m \le q^2+1$), by the same proof.

Corollary 3.2 [new; Theorem when the hub graph exists]. With full slots ($K_1 = K_{\mathcal{B}}$, $K_2 = K_{\mathcal{A}}$) Theorem 3.1 gives $n_b(q+1+k,7) \ge F_k(q)$, where

$$F_k(q) = 2(2k+1)(q+1)(q^4+q^2+1) + \begin{cases} k\,(q^4+2q^2+q+1), & k \text{ even}, \\ \frac{k-1}{2}\,(2q^4+q^3+q^2+2q+1) + (k+1)(q^2+1), & k \text{ odd}. \end{cases}$$

In particular

$$F_2(q) = 10N(q)+2q^4+4q^2+2q+2, \quad F_3(q) = 14N(q)+2q^4+q^3+5q^2+2q+5, \quad F_4(q) = 18N(q)+4q^4+8q^2+4q+4.$$

The gain over the level-graph value $2(2k+1)N(q)$ (the 2013 table value when $q+1+k \le 16$) is $kq^4+O(q^2)$ for k even and $(k-1)q^4+\frac{k-1}{2}q^3+O(q^2)$ for k odd; the earlier general statement, `D7_HEXAGON.md` Corollary 9.3 ($k=2$), gave $2(q+2)(q^2+1) \approx 2q^3$, and none was stated there for $k \ge 3$ except for $q = 3^h$ (Corollary 9.2). With a larger non-friendly set m the formula changes by $k(m-q^2-1)$ for k even and $(k+1)(m-q^2-1)$ for k odd.

Computer-checked (`d7thm/cells.py`). The closed forms agree with the slot formulas for $q \le 37$, $k \le 6$.

4. Hub graphs

The bound needs $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; q-k)$ with $K \approx \lfloor k/2 \rfloor q^4$ hubs per side and degree $q-k$. The Moore-type bound for same-colour distance 6 is about $(q-k)^6$, so there is room, but the explicit hub graph of `D7_HEXAGON.md` (Lemma 7.1, a plane with pendant trees) has only about $\frac{4}{27}(q-k)^4$ vertices per colour. Write $K_0(s) = (s+1)(s^4+s^2+1)$ for the number of points of a hexagon of order (s,s).

Lemma 4.1 (*blown-up hexagon*) [Theorem; elementary]. Let s be a prime power and $c \ge 1$. Replace every point and every line of $H(s)$ by between 1 and c copies, and join two copies when the originals are incident. The result is an $\mathrm{HG}(K_1, K_2; c(s+1))$ for any chosen $K_1, K_2 \in [K_0(s), cK_0(s)]$, and any two of its vertices of different colours are at distance at most 5.

Proof. A path between copies of $x \ne y$ projects to a walk from x to y of the same length, and a geodesic from x to y lifts to copies, so the distance is $d(x,y)$; two copies of one element are at distance 2. A copy of x has degree $\sum_{y \text{ I } x} t_y \le c(s+1)$. □

Lemma 4.2 (*sums*) [new; Theorem]. Let $Y_1, \dots, Y_r$ be bipartite graphs with colour classes (A_i, B_i), each with all same-colour distances at most 6, and for $i < r$ also all different-colour distances at most 5. Suppose $|A_j| \le |B_i|$ and $|B_j| \le |A_i|$ for $i < j$. For every $i < j$ add an injective map $A_j \to B_i$ and an injective map $B_j \to A_i$, as edges. The result is an $\mathrm{HG}(\sum |A_i|, \sum |B_i|;\ \max_i \Delta(Y_i) + r - 1)$.

Proof. The new edges join the colours A and B, so the graph is bipartite with classes $\bigcup A_i$, $\bigcup B_i$. A vertex of Y_i gains one edge to each Y_j, $j < i$, and at most one from each Y_j, $j > i$. Same-colour pairs inside one Y_i are at distance at most 6 in Y_i. For $x \in A_j$ and $z \in A_i$ with $i < j$, x has a neighbour $b \in B_i$, and $d_{Y_i}(b,z) \le 5$ since $i < r$; so $d(x,z) \le 6$. The same for B. □

Lemma 4.3 (`D7_HEXAGON.md`, Lemma 7.1, with the colours sized separately) [restatement; Theorem]. Let s be a prime power, $n = s^2+s+1$, $t = \delta - s - 1 \ge 0$, $u = \delta - 1$. Attach to the incidence graph of $\mathrm{PG}(2,s)$ c_P pendant vertices at the points and c_L at the lines (at most t per core vertex), and g_P, g_L pendant vertices below these (at most u per vertex). This is an $\mathrm{HG}(n+c_L+g_P,\ n+c_P+g_L;\ \delta)$. *Proof.* The distance argument of Lemma 7.1 does not use the numbers of pendant vertices. □ (Different-colour distances can be 7 here, so such a part can only be the last part Y_r in Lemma 4.2; squad card c5 confirmed this with a negative control.)

Theorem 4.4 [new; Theorem; (b) computer-assisted: the decompositions are found and verified by a program]. Let $k \ge 2$. An $\mathrm{HG}(K_{\mathcal{B}}(q,k), K_{\mathcal{A}}(q,k); q-k)$ (full slots of Theorem 2.2, $m = q^2+1$) exists

(a) for every prime power $q \geq 300k$;

(b) for $2 \leq k \leq 12$, for every prime power $q \geq q_k$, where

k	2	3	4	5	6	7	8	9	10	11	12
q_k	16	23	23	29	23	31	29	37	32	43	41

These are the thresholds of our search (`d7thm/hubopt.py`), not of the method; they are not monotone in k.

Proof. (a) Let $q \geq 300k$ (so $q \geq 600$). The slot counts satisfy $q^4 < \Lambda \leq K_{\mathcal{A}} \leq K_{\mathcal{B}}$, $K_{\mathcal{B}}/K_{\mathcal{A}} \leq P/\Lambda \leq 1 + 1/q$ and $K_{\mathcal{B}} \leq kP \leq k(q+1)^4$. Put $x = K_{\mathcal{A}}^{1/5} - 1$ ($> q^{4/5} - 1 > 160$). By Nagura's theorem [Na52] (for $y \geq 25$ there is a prime in $(y, \frac{6}{5}y)$) there is a prime s with $x/1.2 < s < x$. Then $K_0(s) < K_0(x) < (x+1)^5 = K_{\mathcal{A}}$, and, with $K_0(y) \geq y^5$, $x^5 \geq K_{\mathcal{A}}(1 - 5K_{\mathcal{A}}^{-1/5})$ (Bernoulli) and $K_{\mathcal{A}}^{-1/5} < q^{-4/5} \leq 0.006$,

$$\frac{K_{\mathcal{B}}}{K_0(s)} < \frac{1.2^5 K_{\mathcal{B}}}{x^5} \leq \frac{2.4884\,(1 + 1/q)}{1 - 0.03} < 3.$$

So Lemma 4.1 with $c = 3$ gives an $\mathrm{HG}(K_{\mathcal{B}}, K_{\mathcal{A}}; 3(s+1))$, and $3(s+1) < 3(x+1) = 3K_{\mathcal{A}}^{1/5} \leq 3k^{1/5}(q+1)^{4/5} = 3(q+1)(k/(q+1))^{1/5} \leq 3 \cdot 300^{-1/5}(q+1) < 0.959\,(q+1) \leq q - k$, the last step because $0.041(q+1) \geq 12.3k \geq k+1$. (b) For $q \leq 200$ the program `d7thm/hubopt.py` (an enumeration of multisets of parts with bounds) writes for each (q, k) a decomposition into parts of Lemmas 4.1–4.3, combined by Lemma 4.2, and verifies it (sizes add up, the order condition, the degrees, the pendant-tree parameters, which `hubopt.out` prints); each line can be checked by hand. Examples: $(q, k) = (16, 2)$, $K = 66{,}065$: $H(8) + H(7) + H(5)$ and two copies of $H(4)$ blown up to 2,730 and 2,372 (degrees $9+4$, $8+4$, $6+4$, $10+4 \leq 14$); another one, built and checked below, is $H(8) + H(7) + H(5) + H(5)$ and a pendant-tree part on $\mathrm{PG}(2, 5)$ with 124 and 116 vertices at depth 1 and 1,049 and 1,041 at depth 2 (1,196 per colour, degree 10). $(q, k) = (23, 3)$: $H(11) + H(9) + H(8)$, the last blown up to 49,506 and 37,867. Below the full slots the same search gives the hub graphs of §5.2, for instance $(13, 4)$: $H(7) + H(7)$, two hexagons of order 7 joined by a perfect matching from the points of each to the lines of the other (39,216 per colour, degree $8 + 1 = 9$). For $200 < q \leq 2 \cdot 10^5$ Lemma 4.1 alone suffices (`d7thm/hubcap.py`, which finds s and c for every such q); for $q > 2 \cdot 10^5 \geq 300k$ use (a). □

Computer-checked. `d7thm/build_cert.py` builds the hub graph of a certificate (hexagons from `research/d10D7/hexgen2.c`, random injections) and `hgcheck edges` checks it: $(q, k) = (16, 2)$, $\mathrm{HG}(66065, 66065; 14)$ (the one behind $(19, 7)$), $(17, 2)$, $\mathrm{HG}(84117, 84117; 15)$, and $(23, 3)$, $\mathrm{HG}(293092, 281453; 20)$: no same-colour pair at distance more than 6, degrees as claimed (`d7thm/cert_checks.out`).

Computer-checked hub graphs below q_k (`d7thm/hgcheck.c`, own code: bit-parallel breadth-first search from every hub; `d7thm/hgcheck_lab.out`, `d7thm/hgrun.out`). The lab's stored hub graphs, read from the solution files, re-checked standalone: $\mathrm{HG}(15541, 15541; 9)$ [(14,7)], $\mathrm{HG}(29535, 29535; 11)$ [(16,7)], $\mathrm{HG}(32356, 30327; 10)$ and $\mathrm{HG}(32356, 30158; 10)$ [(17,7)], $\mathrm{HG}(25308, 25308; 9)$ [(18,7), PSL(2,37)], $\mathrm{HG}(66065, 66065; 14)$ [(19,7)], $\mathrm{HG}(70163, 66322; 13)$ [(20,7)], and the Haar graph over $\mathrm{PSL}(2, 29)$ of job "conics", $\mathrm{HG}(12180, 12180; 8)$ [(15,7), `conics/d15D7/hub_psl29.txt`]: all VALID. The two cluster-assisted ones [(15,7) 16,152, (18,7) 34,440] are not hub graphs in this sense (their same-colour pairs are joined through the levels; in the hub graph alone 452,256 and 654,360 ordered same-colour pairs are at distance more than 6, the second number as in `conics2/README.md`); for them (HG) is covered by the all-pairs checks of the whole graphs. New random graphs (`hgcheck rand`, up to four seeds, aimed 2-switch repair when $K \leq 60{,}000$) at the slot counts of Theorem 2.2 with $m = q^2 + 1$, for the (q, k) with $11 \leq q < q_k$, $k \leq 5$ (ordered same-colour pairs at distance more than 6 are "far pairs"; Haar graphs of $\mathbb{Z}_K$ were also tried and never covered):

(q, k)	$\delta = q - k$	$K_{\mathcal{B}}$	$K_{\mathcal{A}}$	$M(\delta)/K_{\mathcal{B}}$	result
$(11, 2)$	9	14,895	14,895	20.1	found (rand seed 2)
$(13, 2)$	11	28,913	28,913	38.4	found (rand seed 1)

(q,k)	$\delta = q-k$	$K_{\mathcal{B}}$	$K_{\mathcal{A}}$	$M(\delta)/K_{\mathcal{B}}$	result
$(11,3)$	8	16,228	15,017	8.5	not found (last graph tried: 602,324 far pairs)
$(13,3)$	10	31,112	29,083	19.2	not found (last graph tried: 44 far pairs)
$(16,3)$	13	70,163	66,322	46.4	found (rand seed 1)
$(17,3)$	14	89,032	84,407	58.7	found (rand seed 1)
$(19,3)$	16	137,924	131,425	88.5	found (rand seed 1)
$(11,4)$	7	29,790	29,790	1.9	not found (last graph tried: 270,820,464 far pairs)
$(13,4)$	9	57,826	57,826	5.2	not found (last graph tried: 37,527,036 far pairs)
$(16,4)$	12	132,130	132,130	14.7	not found (last graph tried: 13,824 far pairs)
$(17,4)$	13	168,234	168,234	19.4	not found (last graph tried: 212 far pairs)
$(19,4)$	15	262,126	262,126	30.9	found (rand seed 1)
$(11,5)$	6	32,334	29,912	0.6	not found (last graph tried: 1,190,378,570 far pairs)
$(13,5)$	8	62,054	57,996	2.2	not found (last graph tried: 1,152,049,654 far pairs)
$(16,5)$	11	140,069	132,387	7.9	not found (last graph tried: 43,492,232 far pairs)
$(17,5)$	12	177,774	168,524	11.0	not found (last graph tried: 5,026,754 far pairs)
$(19,5)$	14	275,486	262,488	19.0	not found (last graph tried: 11,012 far pairs)

We did not run $k = 5$ with $q \geq 23$ (more than 300,000 hubs per colour) or $k = 6$.

Heuristic. Let $M(\delta) = 1 + \delta(\delta-1) + \delta(\delta-1)^3 + \delta(\delta-1)^5$, the most vertices of one colour within distance 6 of a vertex in a graph of maximum degree δ. In the table above a random hub graph worked exactly when $M(\delta)/K_{\mathcal{B}} \geq 20$; at 19.0–19.4 ($(19,5)$, $(13,3)$, $(17,4)$) tens to thousands of far pairs were left, and below that random graphs are far off. Where they fail the lab used Haar graphs over $\mathrm{PSL}(2,p)$ and paths through the levels, which reach fewer hubs than the slots (16,152 of 16,309 at $(15,7)$, 34,440 of 59,070 at $(18,7)$).

5. The cells

5.1 The fifteen built graphs

`d7thm/cells.py` recomputes, for every graph built in the jobs "conics" and "conics-2", the slot vectors of §2 from q, the type string, $|\mathcal{P}_S| = P(q)$, $|R_L|$ (q^4+q without S_0, q^4+q^2+q with it) and m, and the order $2(2k+1)N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}$ from the hub counts. **Computer-checked**: all fifteen agree with the builders (`cells.out`): every slot vector equals the one printed in the make logs on branch `claude/conics-2-diameter-7-ic0oat`, and every order equals the n of the solution file. The current graphs:

cell	q, types	m	hubs per side	hub graph	n	all-pairs check	the bound follows from
(14,7)	11, $HDHDH$	768	15,541 + 15,541	random	1,802,642	passed	Thm 3.1 + computer (set, hub graph)
(15,7)	11, $HHDHHDD$	768	16,152 + 16,152	Haar, clusters	2,512,488	passed	all-pairs check
(16,7)	13, $HDHDH$	792	29,535 + 29,535	random	4,081,410	passed	Thm 3.1 + computer (set, hub graph)
(17,7)	13, $HHDHHDD$	792	32,356 + 30,327	random	5,693,959	passed	Thm 3.1 + computer (set, hub graph)
(18,7)	13, $HHDHDHDHD$	792	34,440 + 34,440	Haar, clusters	7,309,092	passed	all-pairs check
(19,7)	16, $HDHDH$	257	66,065 + 66,065	random	11,316,940	9/16 ranges	**Theorem** ($= F_2(16)$)
(20,7)	16, $HHDHHDD$	257	70,163 + 66,322	random	15,795,219	9/24 ranges	Thm 3.1 + computer (hub graph) ($= F_3(16)$)

All use $\Lambda = q^4+q^2+q$ (with S_0). Hub graphs: "random" means random bipartite of degree $q-k$ (checked standalone, §4); "Haar, clusters" means a Haar graph over $\mathbb{Z}_{673} \rtimes \mathbb{Z}_{24}$ resp. $\mathrm{PSL}(2,41)$ whose same-colour pairs are partly joined through the levels (`conics2/README.md` §5). The all-pairs states are those of `conics2/README.md` at 11:44 UTC.

At $(19,7)$ and $(20,7)$ the built graphs use exactly the sets of Theorem 3.1 ($m = q^2+1$, the elliptic section), so their orders are $F_2(16)$ and $F_3(16)$. For $(19,7)$ every ingredient is proved: the sets (Corollary 1.8, Proposition 1.9), the repair of every transition (Theorems B and C), the slot counts (Theorem 2.2) and an $\mathrm{HG}(66065, 66065; 14)$ (Theorem 4.4: $H(8)+H(7)+H(5)+H(5)$ and a pendant-tree part, joined by Lemma 4.2). So

$$n_b(19,7) \;\geq\; 11{,}316{,}940$$

holds by proof, independently of the all-pairs check of the built graph (which uses a random hub graph and was not complete when job "conics-2" ended). For $(20,7)$ the hub graph $\mathrm{HG}(70163, 66322; 13)$ is not covered by Theorem 4.4; it is computer-checked (the builder's graph, re-checked standalone by `hgcheck`, and a second random graph, seed 1, `hgrun.out`); with the proved equal-sided hub graph of 65,737 hubs per colour the bound is 15,790,208. This is a statement about the theorems, not a record claim: the lab records values after the coordinator's own check.

A larger (18,7) graph [new; Theorem, built and partly computer-checked]. At $(q,k) = (13,4)$ the cluster-assisted Haar hub graph of the current build has 34,440 hubs per colour, while Lemma 4.2 gives

HG(39216, 39216; 9) ($H(7) + H(7)$, §4). With the build's sets (slots 59,070 per colour) or those of Theorem 3.1 (slots 57,826), Theorem 3.1 gives

$$n_b(18, 7) \;\geq\; 18 \cdot 402{,}234 + 2 \cdot 39{,}216 \;=\; 7{,}318{,}644,$$

9,552 more than the lab's 7,309,092. We built it with `conics2/typed_build.py` (same hexagon file, types $HHDHDHDHD$, the 792-point mirrored set, S_0 included; the hub graph from `d7thm/build_cert.py`): the builder re-checks every hypothesis (segment and dual segment sets, the local duality at all 792 points, same-colour hub distances at most 6, degrees 15 to 18) and writes n = 7,318,644, 65,612,918 edges, edge-list sha256 `c80d9c0e6a569508b2314c3c3cd6c53babce0ecb844017fdf7f86f297426ed1a`; the solution file `research/frontier/sols/d18D7_hex13_typedwhisk_h7h7_n7318644.sol` rebuilds it byte for byte (`typed_build.py build`). **Computer-checked:** (i) its level part (all edges between level vertices, 64,554,086 of them) is identical to that of the lab's 7,309,092 graph, rebuilt from its solution file (sha256 `314d5db9…`, as recorded); (ii) `apsp_fast` (`conics/apsp_fast.c`) from all 78,432 hubs: bipartite, degrees 15 to 18, every hub within distance 7 of every vertex (`RESULT VALID d=18 D=7 n=7318644` for the sources 7,240,212 to 7,318,643); LEVELCHECK Pairs of level vertices are within distance 7 inside two levels by Theorems B and C, whose hypotheses the builder checked; a full all-pairs check (about 20 hours on 4 cores here) was not run and is left to the coordinator (`d7thm/d18D7/`).

5.2 The cells $\Delta = 14, \ldots, 26$

For each Δ, the best (q, k) with $q + 1 + k = \Delta$; "level graphs" is $\max 2(2k+1)N(q)$, the 2013 table value for $\Delta \leq 16$ and the lab's beyond-table value for $\Delta = 17, \ldots, 20$ (`research/frontier/beyond/BEYOND_TABLE.md`, branch `claude/new-cells-beyond-etos3t`). $F_k(q)$ is Corollary 3.2 with $m = q^2 + 1$; its hub graph is "proof" (Theorem 4.4), "computer" (`hgcheck`) or "open". "Proved" is the value of Theorem 3.1 with the hub graph that `d7thm/hubopt.py` finds below the slots (a sum by Lemma 4.2, shown in the last column; full slots where Theorem 4.4 applies). The search is not proved optimal, so these are values found, not the best the lemmas allow.

cell	level graphs	(q, k)	$F_k(q)$	hub graph at full slots	proved	at (q, k): hubs	hub graph of the proved value
(14,7)	1,771,560	$(11, 2)$	1,801,350	computer	1,797,726	$(11, 2)$: 13,083 + 13,083	H(5) + H(5) + H(5) + H(4)
(15,7)	2,480,184	$(11, 3)$	2,511,429	open	2,503,620	$(11, 3)$: 11,718 + 11,718	H(5) + H(5) + H(5)
(16,7)	4,022,340	$(13, 2)$	4,080,166	computer	4,079,910	$(13, 2)$: 28,785 + 28,785	H(7) + H(5) + H(5) + H(4)
(17,7)	5,631,276	$(13, 3)$	5,691,471	open	5,686,116	$(13, 3)$: 27,420 + 27,420	H(7) + H(5) + H(5)
(18,7)	7,240,212	$(13, 4)$	7,355,864	open	7,318,644	$(13, 4)$: 39,216 + 39,216	H(7) + H(7)
(19,7)	11,184,810	$(16, 2)$	11,316,940	proof	11,316,940	$(16, 2)$: 66,065 + 66,065	H(8) + H(7) + H(5) + H(4)x2 + H(4)x2
(20,7)	15,658,734	$(16, 3)$	15,795,219	computer	15,791,202	$(16, 3)$: 66,234 + 66,234	H(8) + H(7) + H(5) + H(5) + H(4)
(21,7)	21,120,372	$(17, 3)$	21,293,811	computer	21,290,219	$(17, 3)$: 85,440 + 84,407	H(8) + H(8) + H(5) + H(5) + H(4)x2
(22,7)	27,154,764	$(17, 4)$	27,491,232	open	27,476,636	$(17, 4)$: 160,936 + 160,936	H(9) + H(8) + H(8) + H(7)
(23,7)	36,591,240	$(19, 3)$	36,860,589	computer	36,857,870	$(19, 3)$: 135,205 + 131,425	H(9) + H(8) + H(7) + H(5)x2 + H(5)
(24,7)	47,045,880	$(19, 4)$	47,570,132	computer	47,566,386	$(19, 4)$: 260,253 + 260,253	H(9) + H(9) + H(9) + H(8) + H(7) + H(5)

cell	level graphs	(q,k)	$F_k(q)$	hub graph at full slots	proved	at (q,k): hubs	hub graph of the proved value
(25,7)	57,500,520	$(19,5)$	58,038,494	open	58,013,214	$(19,5)$: 256,347 + 256,347	H(9) + H(9) + H(9) + H(8) + H(7)
(26,7)	67,955,160	$(19,6)$	68,741,538	open	68,663,784	$(19,6)$: 354,312 + 354,312	H(11) + H(11)

At $(15,7)$ the lab's Haar graph over PSL$(2,29)$ (`CONICS.md` §3.4; re-checked above: same-colour distances at most 6 in the hub graph alone) fits the slots of Theorem 2.2 with $m = q^2+1$ as well, so Theorem 3.1 with that computer-checked hub graph gives $14N(11) + 2 \cdot 12{,}180 = 2{,}504{,}544$.

At $(18,7)$ the proved value, 7,318,644, exceeds the lab's built 7,309,092 (§5.1). The proved values exceed the level-graph values at every cell $\Delta = 14, \ldots, 26$. For $k \geq 3$ the lab's notes had no general statement except for $q = 3^h$ (`D7_HEXAGON.md`, Corollary 9.2; §9 there says none covers $(15,7)$). The values with computer-found ingredients (the lab's sets $m = 768, 792$ and hub graphs) are the built values of §5.1 and are larger at $q = 11, 13$.

6. Summary: what is new

result	label	status
Theorem 1.4: lines meeting v, whiskers and the spread lines missed by them form a dual segment set, for any hexagon, spread and whisker set; maximality criterion	new in this form (special cases: `CONICS.md` Lemma 3.1, `conics2` Prop. 1(c),(d))	Theorem, coordinate-free
Proposition 1.5: in a dual segment set whose lines meet only in spread points, at most q^4+q lines are off the spread; $\mathcal{L}_v \cup W$ attains it	new	Theorem
Lemma 1.6: an exact whisker set misses at most q^3-q^2 spread lines, exactly q^2 under the "all or nothing" property	new	Theorem
Theorem 1.7, Corollary 1.8: $\lvert R_H\rvert = (q+1)(q^3+1)$, $\lvert R_L\rvert = q^4+q^2+q$, R_L maximal, every odd q and even $q \geq 4$	restatement of `conics2` Prop. 1 with proof	Theorem; Computer-checked $q = 3, \ldots, 16$ (squad c1–c3, builder; $q = 2$ checked to have no zero-free G)
Theorem 2.1: for every regular tournament and typing, the hub counts per side are at most $\lfloor k/2 \rfloor$ friendly sets, with the exact second-order term	new	Theorem; Computer-checked (all tournaments on ≤ 7 levels, SAT to 13 levels, squad c6)
Theorem 2.2: the alternating layout attains Theorem 2.1 for every k; closed forms	new	Theorem; Computer-checked $k \leq 40$
Theorem 3.1, Corollary 3.2: $n_b(q+1+k,7) \geq F_k(q)$ given the hub graph	new (for $k=2$ it improves `D7_HEXAGON.md` Cor. 9.3 from $\approx 2q^3$ to $\approx 2q^4$ above the level graph; nothing was stated for $k \geq 3$, $q \neq 3^h$)	Theorem

result	label	status
Lemma 4.2: sums of hub graphs; Lemma 4.1: blow-ups; Lemma 4.3: `D7_HEXAGON.md` Lemma 7.1 with the colours sized separately	4.2 new; 4.1 elementary; 4.3 restatement	Theorem; Computer-checked (squad c5)
Theorem 4.4: the hub graphs at full slots exist for $q \geq q_k$ ($q_2 = 16$, $q_3 = q_4 = 23$, ..., $q_k \leq 300k$)	new	Theorem; (b) computer-assisted (certificates by program for $q \leq 200$)
$n_b(19, 7) \geq 11{,}316{,}940$ and $n_b(18, 7) \geq 7{,}318{,}644$ (above the lab's 7,309,092) by proof; proved values above the level graphs at $(14, 7)$–$(26, 7)$ (§5.2)	new consequence	Theorem; the $(18, 7)$ graph built, checked from every hub, full all-pairs check open
The formulas reproduce all fifteen built graphs; the lab's hub graphs re-checked standalone; new random hub graphs for $q < q_k$	—	Computer-checked
Random hub graphs fail at $(q, k) = (11, 3), (13, 4)$; larger dual segment sets than $q^4 + q^2 + q$ for $q \geq 5$; a coordinate-free description of the q^2 lines with $\nu = 0$	—	open; Heuristic remarks in §1.3 and §4

The theorems explain the values built by the two jobs (at $q = 16$ exactly) and give the same kind of bound for every q and every k, with an explicit hub graph wherever $q \geq q_k$. One value is raised: at $(18, 7)$ the explicit hub graph beats the computer-found one, giving 7,318,644 (a candidate until the coordinator's all-pairs check).

Review. A separate referee session re-derived the numerical steps (the Nagura chain, the closed forms, every table entry, all certificates of `hubopt.out` with its own checker, and Theorem 2.1 on all regular tournaments with at most 7 levels) and found no error in the proofs; it found that the first certificate search was far from optimal (now `hubopt.py`, which produced the $(18, 7)$ hub graph) and several wording problems, all corrected here.

Appendix: files

In `research/synthetic/d7thm/` (Python 3; `pip install python-sat` for `slots.py`; C with OpenMP for `hgcheck.c`; commands in `d7thm/README.md`):

- `slots.py` → `slots.out`: §2 (all typings of the rotational tournament for $k \leq 8$, all regular tournaments on 5 and 7 levels, SAT for 9–13 levels).
- `hubcap_lib.py`, `hubcap.py` → `hubcap.out`: Lemma 4.1 coverage for $q \leq 2 \cdot 10^5$ and the inequalities of Theorem 4.4(a); `hubopt.py` → `hubopt.out`: the certificates of Theorem 4.4(b) and of §5.2 for $q \leq 200$ (one line per (q, k), pendant-tree parameters included); `build_cert.py`: the hub graph of a certificate as an edge list.
- `d18D7/`: the $(18, 7)$ graph of §5.1 (builder log, level-part comparison, the check from every hub); its solution file is `research/frontier/sols/d18D7_hex13_typedwhisk_h7h7_n7318644.sol`.
- `hgcheck.c`: standalone check of a hub graph (same-colour distances by bit-parallel BFS from every vertex); modes `sol` (read a solution file), `rand` (random bipartite graph, optional aimed 2-switch repair), `haar` (Haar graphs of $\mathbb{Z}_K$). `hgcheck_lab.out`: the lab's hub graphs; `hgrun.py` → `hgrun.out`: random hub graphs at the slot counts of Theorem 2.2 for $q < q_k$.
- `cells.py` → `cells.out`: §5.1 and the closed forms of Corollary 3.2; `table.py` → `table.out`: §5.2.

Squad round `research/squads/d7thm-r1/`: `BRIEF.md`, cards `c1`–`c6` (code and logs), `scoreboard.jsonl`; logged in `research/squads/LOG.md`. The PDF `D7_THEOREMS.pdf` is made from this file by pandoc and tectonic (`d7thm/make_pdf.sh`).

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